Lab 7 : Configuration of Switch, VLAN configuration and Inter-VLAN Routing

Objectives:

- To create and configure multiple VLANs on a network switch.
- To configure trunking and assign switch ports to appropriate VLANs.
- To implement and verify Inter-VLAN Routing using a router.

Theory:

Switch Configuration:

A network switch functions at the data link layer and connects multiple devices within a local area network (LAN). It forwards data to the appropriate device by analyzing MAC addresses, which reduces data collisions and enhances overall network efficiency. Switch configuration involves enabling ports, grouping devices into suitable VLANs, and managing traffic flow to ensure the network remains organized and performs effectively.

Virtual Local Area Network (VLAN):

A Virtual Local Area Network (VLAN) is a method that allows a single physical switch to be segmented into several logical networks. Rather than arranging devices according to their physical placement, VLANs group them based on roles, functions, or departments. This method helps reduce unnecessary broadcast traffic, improves network security, and simplifies network management. Each VLAN works as its own independent broadcast domain.

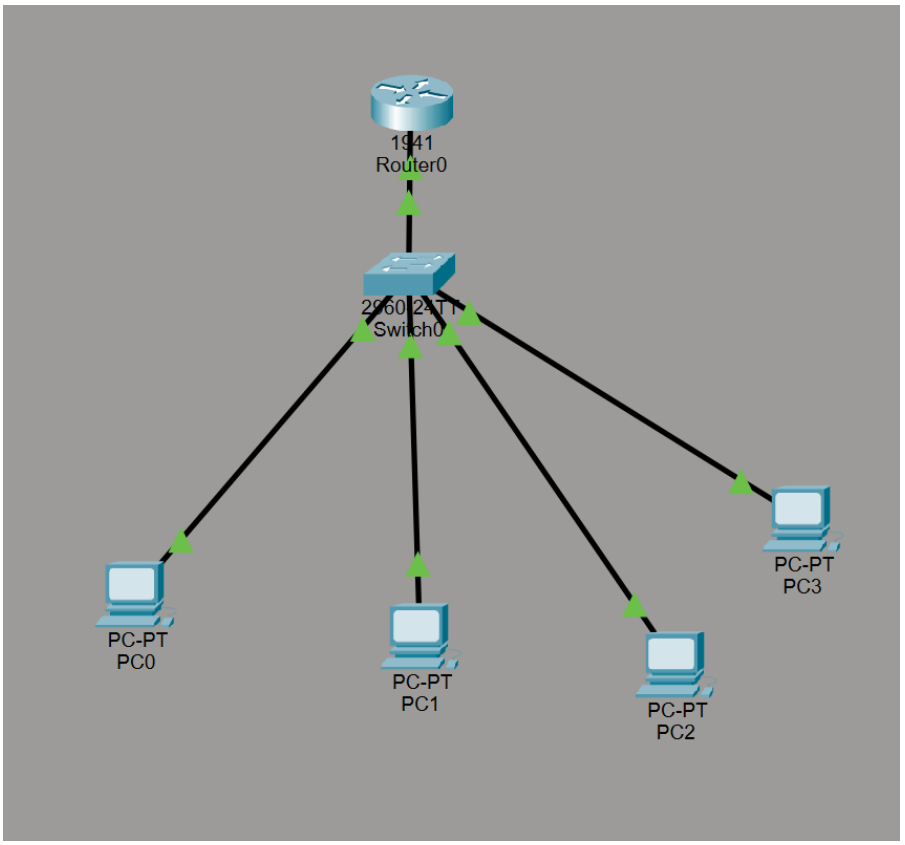
VLAN Configuration:

VLAN configuration starts with creating VLANs by assigning them unique identification numbers and names. Once they are created, switch ports are placed into these VLANs using access mode, allowing the connected devices to belong to the designated network segment. Devices that are part of the same VLAN can communicate directly with each other, while communication between different VLANs requires a routing device.

Inter-VLAN Routing:

Inter-VLAN routing allows communication between different VLANs by using a Layer 3 device, such as a router. In this configuration, the Router-on-a-Stick method is used, where a single physical router interface is split into multiple logical sub-interfaces. Each sub-interface is associated with a specific VLAN and configured with 802.1Q tagging, enabling the router to handle and route traffic between VLANs effectively.

Network Topology :



Observation :

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.2

Pinging 192.168.20.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.20.2: bytes=32 time<1ms TTL=127
Reply from 192.168.20.2: bytes=32 time<1ms TTL=127
Reply from 192.168.20.2: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.20.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Fig: Ping in same LAN

```
C:\>ping 192.168.20.3

Pinging 192.168.20.3 with 32 bytes of data:

Reply from 192.168.20.3: bytes=32 time=1ms TTL=127
Reply from 192.168.20.3: bytes=32 time=3ms TTL=127
Reply from 192.168.20.3: bytes=32 time<1ms TTL=127
Reply from 192.168.20.3: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.20.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 3ms, Average = 1ms
```

Fig: Ping in Different LAN

Configuration :

VLAN and IP Addressing Scheme :

VLAN	Name	Network	Gateway
10	Computer	192.168.10.0/24	192.168.10.1
20	Electronics	192.168.20.0/24	192.168.20.1

Device	IPv4 address	Subnet Mask	Default Gateway
PC0	192.168.10.2	255.255.255.0	192.168.10.1
PC1	192.168.10.3	255.255.255.0	192.168.10.1
PC2	192.168.20.2	255.255.255.0	192.168.20.1
PC3	192.168.20.3	255.255.255.0	192.168.20.1

Commands :

Switch Configuration :

1. **Create VLANS**

   Switch(config)# vlan 10
   Switch(config-vlan)# name
   Computer
   # exit
   Switch(config)# vlan 20

   Switch(config-vlan)# name Electronics

   # exit

2. **Assign port to VLANS**

   Switch(config)# interface range fa0/1-2

   # switchport mode access #
   switchport access vlan 10 #
   exit
   # interface range fa0/3-4
   # switchport mode access
   # switchport access vlan 20
   # exit

3. **Configure Trunk Port (Switch-Router)**

   Switch(config)# interface fa0/5

   # switchport mode trunk
   # exit

Verification Switch :

Switch# show vlan brief

```
Switch>enable
Switch#show vlan brief

VLAN Name                             Status    Ports
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1    default                          active    Fa0/6, Fa0/7, Fa0/8,
Fa0/9
                                                Fa0/10, Fa0/11,
Fa0/12, Fa0/13
                                                Fa0/14, Fa0/15,
Fa0/16, Fa0/17
                                                Fa0/18, Fa0/19,
Fa0/20, Fa0/21
                                                Fa0/22, Fa0/23,
Fa0/24, Gig0/1
                                                Gig0/2
10   Computer                         active    Fa0/1, Fa0/2
20   Electronics                      active    Fa0/3, Fa0/4
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch#
```

Router Configuration (Inter-VLAN Routing) :

Router(config)# interface GigabitEthernet 0/0.10
encapsulation dot1Q 10
ip address 192.168.10.1 255.255.255.0

exit

Router(config)# interface GigabitEthernet 0/0.20
encapsulation dot1Q 20
ip address 192.168.20.1 255.255.255.0

exit

Result:

The switch was successfully configured by creating several VLANs and assigning the appropriate ports to each of them. A trunk link was established to carry VLAN traffic between the switch and the router. Inter-VLAN communication was implemented using router subinterfaces, enabling efficient data transfer both within the same VLAN and between different VLANs.

Discussion & Conclusion:

The network was divided into multiple logical segments using VLANs to reduce broadcast traffic and improve efficiency. Devices were assigned to specific switch ports based on their roles, which helped maintain better organization and control within the network. A trunk link was configured between the switch and the router to carry traffic from multiple VLANs, while inter-VLAN routing was implemented using the Router-on-a-Stick method with router subinterfaces. This setup allowed devices in different VLANs to communicate securely and efficiently. Testing and verification commands confirmed that the VLAN configuration, trunking, and routing were functioning correctly, demonstrating that the approach is reliable for managing segmented networks.